Collaborative Optimization of Morphological Performance in Residential Districts Using Interpretable Machine Learning: a case study area in Changsha

Abstract

Rapid urbanization is reshaping the built environment, making environmental performance a key concern in architectural decision-making. **As residential development shifts toward higher-density forms**, balancing energy efficiency and occupant well-being becomes increasingly difficult. Existing studies often focus on single objectives, limiting their ability to address competing goals (e.g., energy use and thermal comfort), while the gap between advanced optimization methods and design workflows hinders practical adoption. This study compiles a dataset from field surveys and energy simulations of high-rise residential districts and develops predictive models through model benchmarking and hyperparameter tuning. The best-performing model is used as a **surrogate for multi-objective optimization**, achieving **R^2 = 0.91** and substantially improving computational efficiency. An interactive **Streamlit-based decision-support platform with Pareto-front** analysis is further developed to visualize trade-offs between design variables and optimization outcomes, thereby facilitating the translation of optimization results into design practice and **supporting multidimensional sustainable urban planning**.

Technical roadmap

Part 1: Prediction modeling construction

Data generation

Location: Changsha

Data: the districts' morphological parameters

→ **Simulation modeling**

Caculate indicators

- Total energy consumption
- Total incident solar energy
- UTCI

→ **Prediction modeling**

- Machine learning algorithms comparison
- Hyperparameter tuning conducted using Optuna

Part 2: Model interpretability and application

SHAP analysis

SHAP analysis

Importance weights of parameters

→ **Multi-objective optimization**

Python library

pymoo

Algorithm

NSGA-III

→ **Interactive platform**

Python library

Streamlit

Dynamical parameters — Real-time feedback

Research samples selection

To obtain design parameters, a stratified sampling method was employed. Specifically, a representative number of high-rise residential communities were selected from the five major administrative districts of Changsha for field investigation.

Residential communities in Changsha → Administrative stratified sampling

Data acquisition: On-site rearsch — Google Earth

→ Initial samples → Time principle / Spatial principle / Type principle → Final reaserch samples

Time principle: within the period from 2005 to 2025.

Spatial principle: the five districts of Changsha: Furong, Tianxin, Kaifu, Yuelu, and Yuhua.

Type principle: high-rise residential zones.

Morphological parameters and Human-Building-Environment (HBE) indicators

Index	*Name (abbreviation)*	*Formula*	*Introduction*
X1	Enclosure Coefficient (EC)	$EC = \frac{\sum_{i=1}^{n} D_i}{C_{site}}$	the ratio of the total length of residential building façades (D_i) to the perimeter of the site boundary (C_{site}).
X2	Plot Ratio (PR)	$PR = \frac{GFA}{A_{site}}$	the ratio of total above-ground Gross Floor Area (GFA) to the net site area (A_{site}).
X3	Building Density (BD)	$BD = \frac{A_{base}}{A_{site}} \times 100\%$	the ratio of the total building footprint area (A_{base}) to the planned site area (A_{site}).
X4	Orientation of Groups (OG)	$OG = \frac{1}{n}\sum_{i=1}^{n} \theta_i$	the arithmetic mean of the orientation angles (θ_i) for all buildings within the district, where 0° represents due South, with positive values indicating eastward deviation and negative values indicating westward deviation.
X5	Sky View Factor (SVF)	$SVF = 1 - \frac{1}{n}\sum_{i=1}^{n} \frac{H_i}{\sqrt{R^2 - H_i^2}}$	where H_i denotes the building height and R represents the buffer radius.
X6	Openness of Space (OS)	$OS = \frac{A_{site} - A_{base}}{GFA}$	the outdoor open space area per unit of Gross Floor Area, reflecting the potential for site ventilation and hydrothermal environment regulation.
X7	Average Distance (AD)	$AD = \frac{1}{3n}\sum_{i=1}^{n} \sum_{j \in N_i} d(i,j)$	the average distance between each building and its three nearest neighbors (N_i).
X8	Average Floor (AF)	$AF = \frac{GFA}{A_{base}}$	the ratio of Gross Floor Area (GFA) to the total building footprint area (A_{base}).
X9	Avg. Shape Factor (ASF)	$ASF = \frac{\sum(A_{roof} + A_{wall})}{\sum V_i}$	the ratio of the building's heat-dissipating surface area (roof and exterior walls) to the building volume (V_i).
X10	Area Perimeter Ratio (APR)	$APR = \frac{\sum A_{base,i}}{\sum P_i}$	the ratio of the building footprint area (A_{base}) to the sum of the perimeters of the standard floors (P_i).
X11	Avg. Maintenance Coeff. (AMC)	$AMC = \frac{\sum A_{surf,i}}{A_{site}}$	the ratio of the total exterior surface area of all buildings within the plot (A_{surf}) to the site area (A_{site}).

·**Total Energy Consumption (TEC)**

TEC quantifies the integrated annual demand for heating, cooling, and lighting per unit area (kWh/m^2·a), serving as a benchmark indicator to evaluate the collective impact of urban morphology and architectural design on building energy systems.

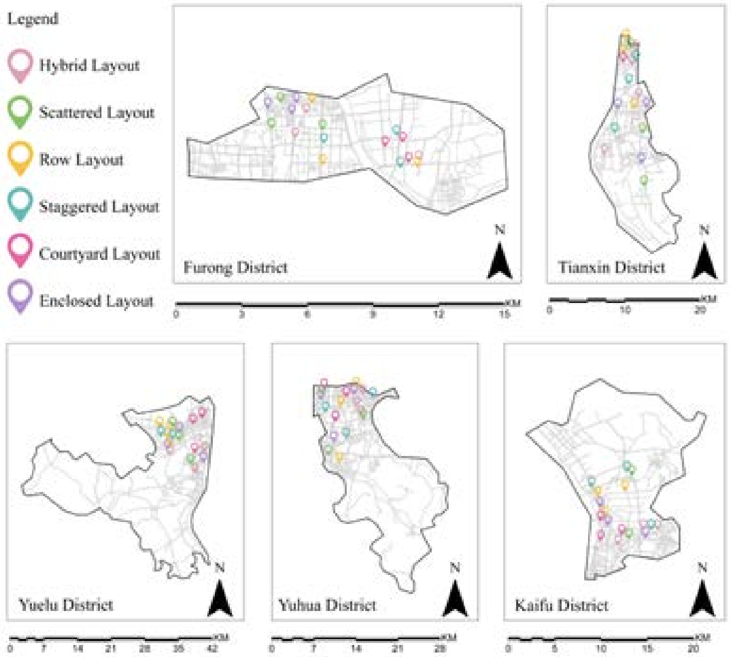
·**Universal Thermal Climate Index (UTCI)**

UTCI provides a comprehensive assessment of pedestrian thermal comfort by integrating air temperature, mean radiant temperature, wind speed, and relative humidity, thereby characterizing the performance of the urban microclimate.

·**Total Incident Solar Exposure (TISE)**

TISE measures the cumulative annual solar radiation received by outdoor surfaces (kWh/m^2·a), serving as a critical metric for evaluating radiation-driven thermal risks and the influence of urban geometry on the external thermal environment.

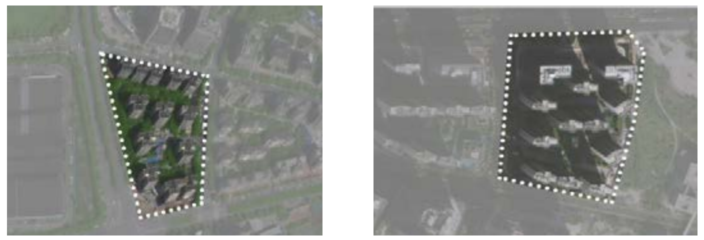
Distribution of the surveyed residential communities



Courtyard layout

Hybrid layout

Scattered layout

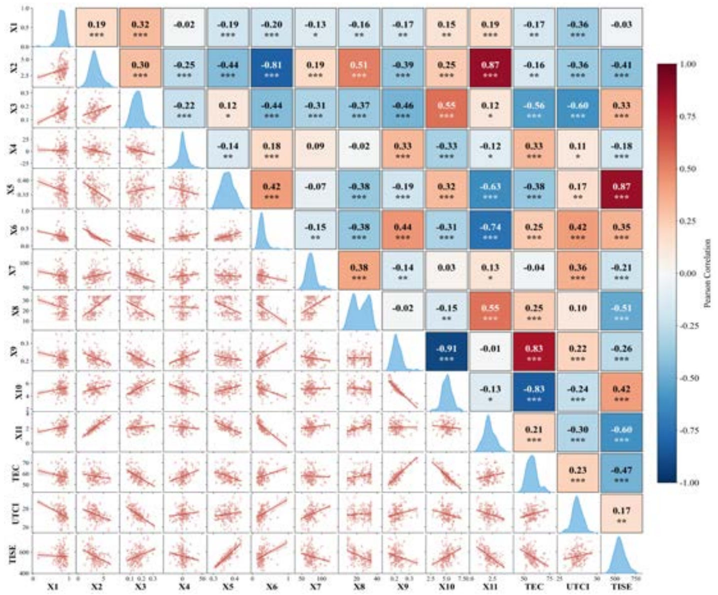


Enclosed layout

Row layout

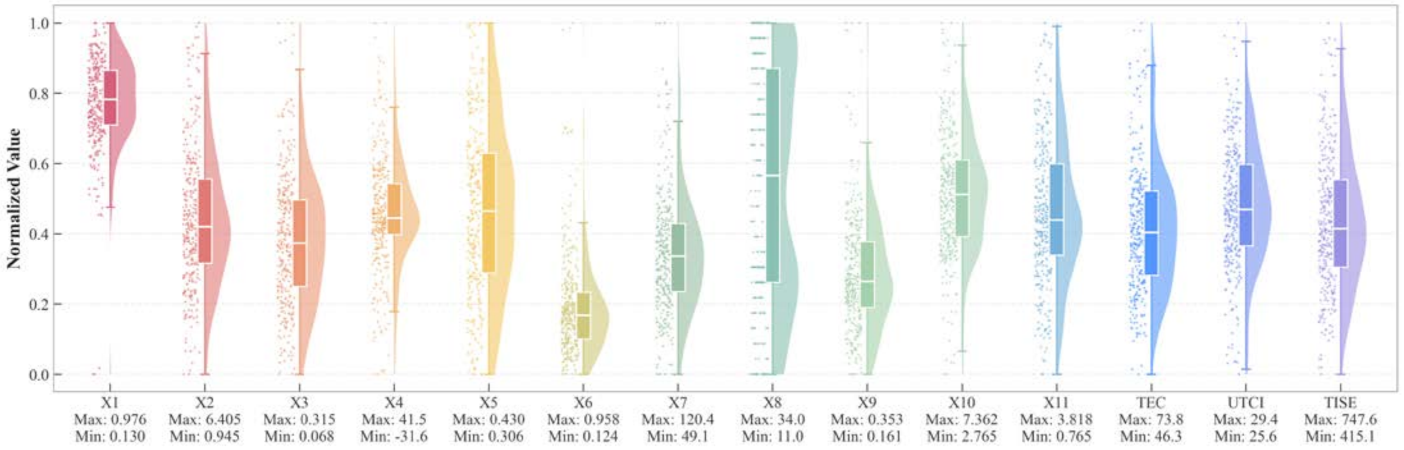
Staggered layout

Analysis of morphological parameters and HBE indicators



Correlation analysis heatmap

The Pearson correlation analysis elucidates the complex interdependence among design parameters, where strong associations (e.g., **r=0.87 between X2 and X10**) reflect the inherent geometric constraints of high-density typologies. While the diagonal density estimations reaffirm the polarized distributions of enclosure (X1) and open space (X6), the significant correlations between specific form factors (X5, X9) and performance metrics (TISE, TEC) underscore the critical impact of morphology on environmental outcomes. This statistical structure confirms that the dataset captures both the geometric diversity and the governing physical mechanisms necessary for robust model training.



Numerical distribution of research sites

Model performance comparison

Representative algorithms

Classical ML: SVR

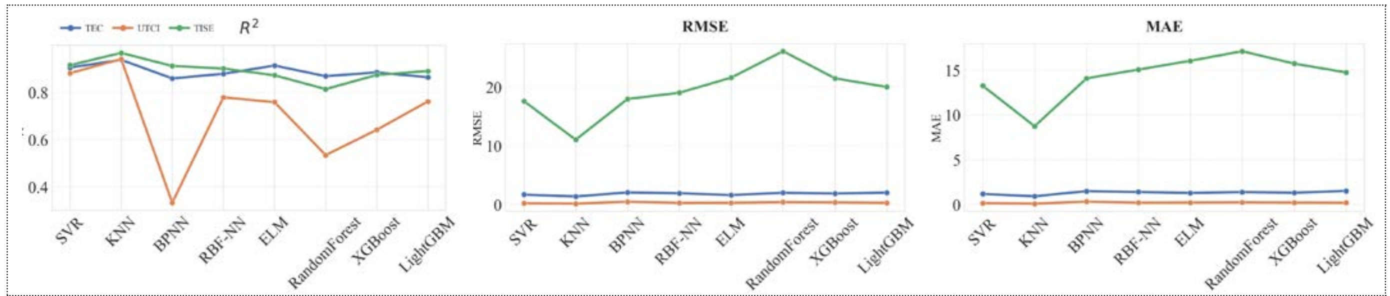
Esembling learning: Random Foresest, XGBoost, LightGBM

Neural networks: BPNN, RBF-NN, ELM

Hyperparameters optimization

- Each algorithms' key parameters
- Using optuna framework
- Tree-structured Parzen Estimator Bayesian optimization

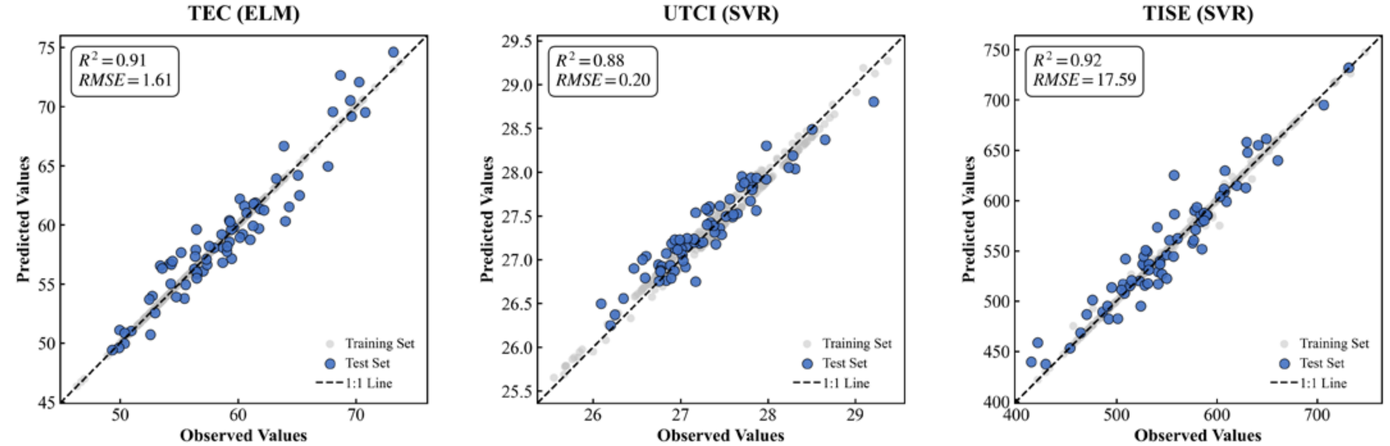
→ **Best models of BPMs and their hyperparameters' combinations**



Comparison of each machine learning model's performance

Predictive performance of BPMs

Prediction vs truth plot visually validates the predictive performance of BPMs, where the tight convergence of scatter points along the 1:1 diagonal signifies robust predictive accuracy across the design space. Specifically, the ELM achieved high-precision fitting for the non-linear trends of **TEC (R2=0.91)**, while SVR captured the sensitivity of **UTCI (R2=0.88)** and the physical consistency of **TISE (R2=0.92)** through global convergence. The alignment of error distributions, optimized via the Optuna-TPE framework, confirms that these surrogates effectively balance bias and variance, providing a computationally efficient and scientifically rigorous foundation for subsequent multi-objective optimization.

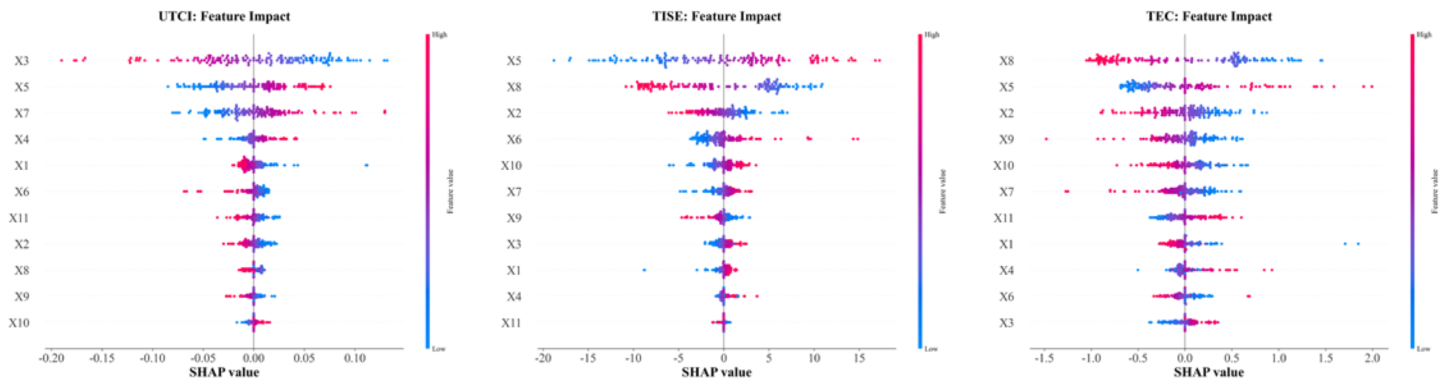


Best hyperparamters	Best hyperparamters	Best hyperparamters
n_hidden: 816	C: 181.54	C: 560
activation: sigmoid	gamma: 0.32	gamma: 0.17

Prediction vs truth plot

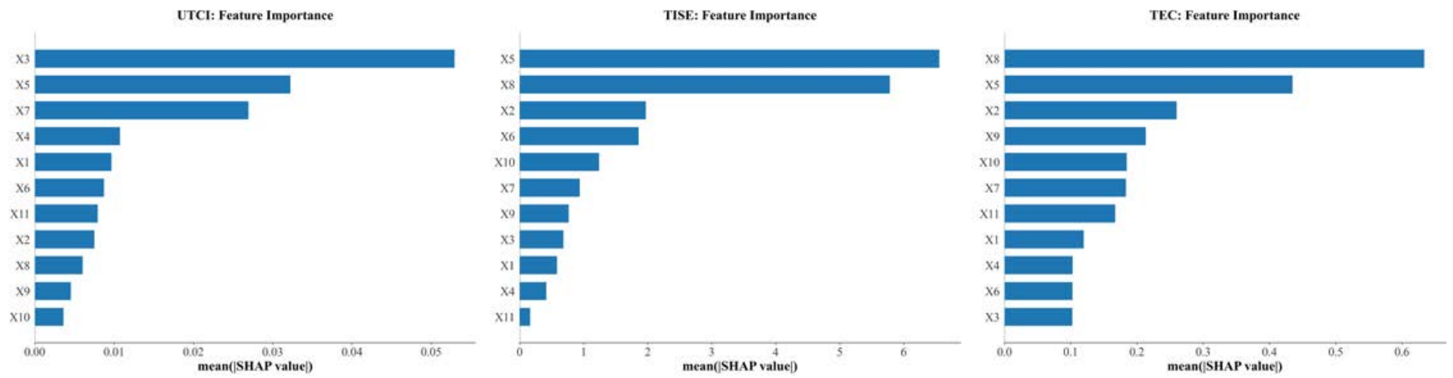
SHAP analysis

Beewarm plot illustrates the feature importance rankings for the three objective functions, while Beewarm plot represents the **feature dependency relationships of the top three morphological parameters for each HBE indicator.**



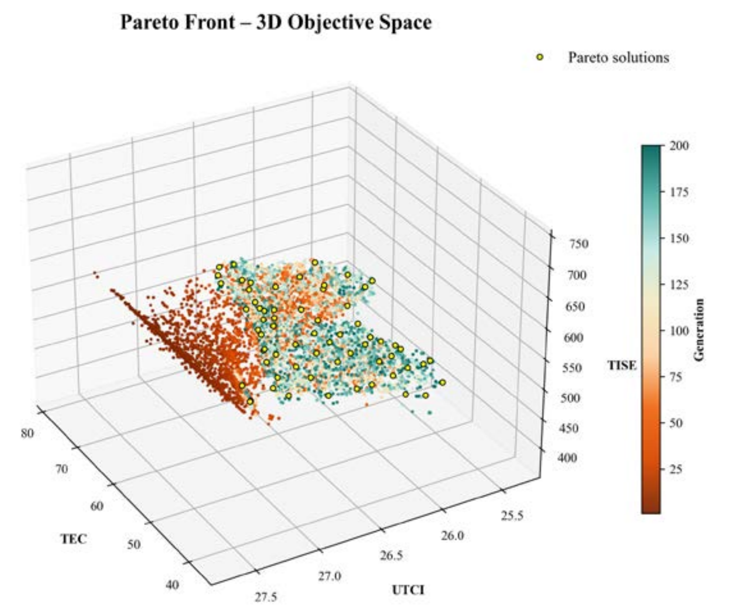
SHAP beewarm plots

In terms of global feature importance, **UTCI is predominantly governed by X3**. This underscores the pivotal role of X3 in regulating the microclimatic thermal environment. Conversely, **TISE exhibits a dependency on X5**, whereas **TEC is primarily driven by X8**, suggesting that energy performance is more sensitive to complex building morphological parameters rather than singular metrics.



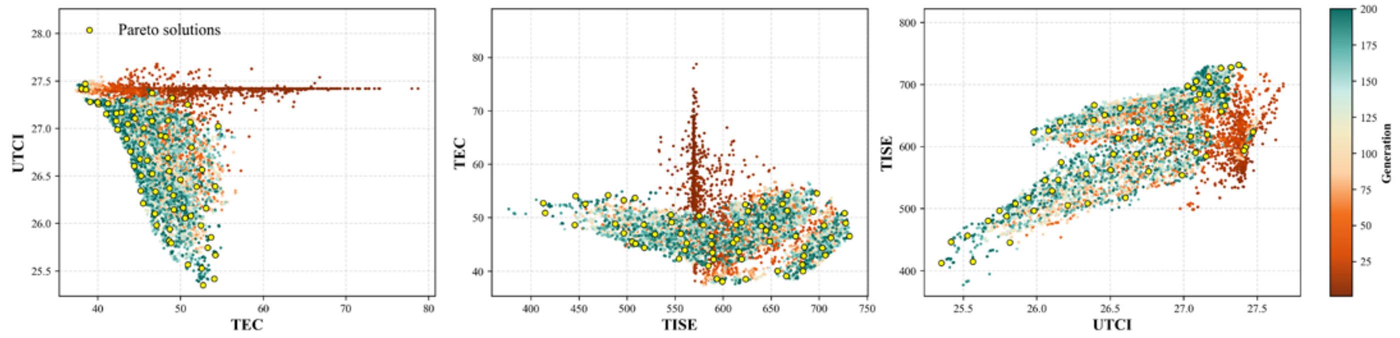
Global feature importance

Multi-objective optimization

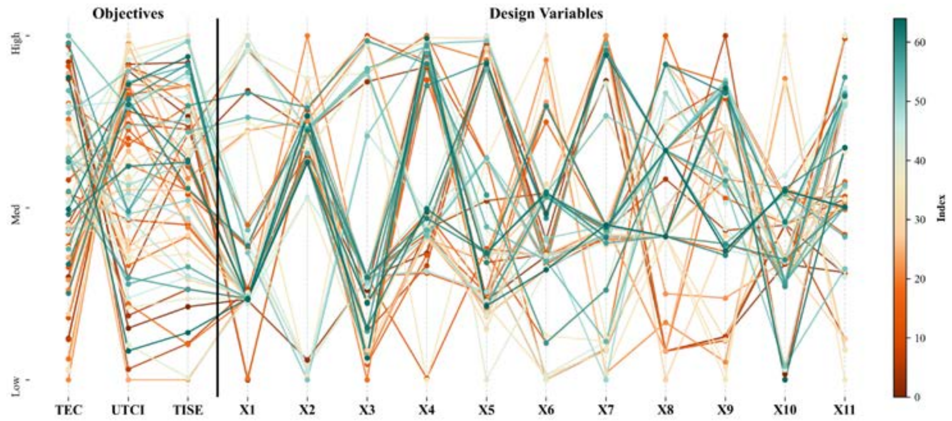


Leveraging the developed BPMs as surrogate models, a Multi-Objective Optimization algorithm based on **NSGA-III** was implemented to perform a **200-generation** evolutionary search across the morphological design space. This process was designed to elucidate the complex trade-off relationships among the HBE indicators. Consequently, the resulting **3D Pareto front** visualizes the optimized spatial distribution spanning the TEC, UTCI, and TISE objectives.

• The non-monotonic TEC-UTCI profile indicates that while synergies exist initially, pushing TEC to minimal extremes compromises outdoor comfort due to heat entrapment in overly compact forms.
• A positive correlation between TISE and UTCI highlights that maximizing solar potential inherently increases thermal stress
• Pursuing higher TISE values correlates with increased TEC, as the larger surface-to-volume ratios required for solar access exacerbate envelope heat losses



Pareto front - 2D objective projections



Pareto front parallel coordinates

Convergence

X2, X3, X4, X9, X9, X10

Critical variables with highly constrained optimal ranges essential for maintaining multi-objective performance.

Divergence

X1, X5, X6, X7, X8, X11

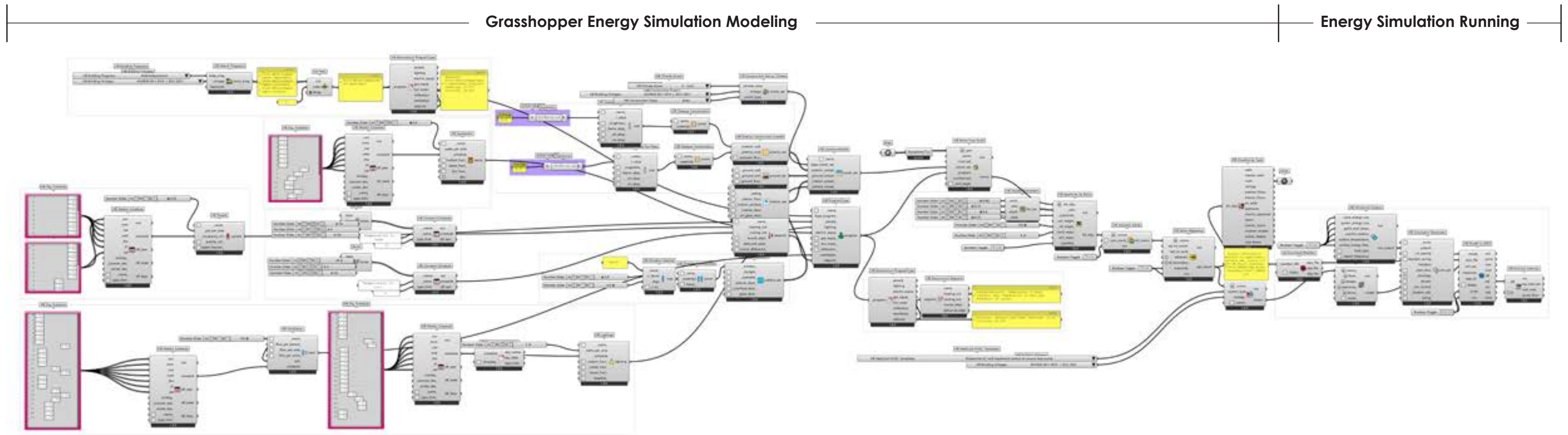
Flexibly accommodate non-performance-related aesthetic or functional requirements with minimal disruption to the overall performance equilibrium.

Streamlit-based decision-support platform

Streamlit platform deployed at
https://changsha-h2o-app-6kksejdp6swqxzagrquqjc.streamlit.app/

Building decision support system was developed. This tool is designed to translate complex, high-dimensional optimization data into an understandable and comparative framework for design practitioners. Functionally, the system accepts user-defined variations in design parameters, maps the resulting scheme within the 3D objective space, and benchmarks its performance relative to the optimal solutions on the Pareto frontier.

Core execution flow



Algorithm: Multi-Target Machine Learning Optimization and Evaluation
Input:
- Feature Matrix: X
- Target Variables: Y (TEC, UTCI, TISE)
- Candidate Models: SVR, KNN, BPNN, RBF-NN, ELM, RF, XGBoost, LightGBM

Parameters:
- Scaling Method: StandardScaler
- Hyperparameter Optimization: Optuna (Bayesian Optimization)
- Optimization Trials: 50 per model
- Evaluation Metrics: R², RMSE, MAE

Output:
- Optimized Predictive Models
- Fitted Global Scaler
- Comprehensive Performance Metrics Report

```
START
 Apply StandardScaler to Feature Matrix X.
 FOR each target IN Y DO:
     Split X_scaled and target into training and testing sets.
     FOR each model IN Candidate Models DO:
         i. Perform Bayesian hyperparameter tuning via Optuna for 50 trials.
         ii. Fit the Final_Model using Best_Params.
         iii. Predict outcomes and compute Evaluation Metrics.
     END FOR
 END FOR
 Export optimized Final_Models.
END
```

Algorithm: Multi-Objective Optimization using NSGA-III
Input:
- Surrogate Models: ELM (TEC), SVR (UTCI), SVR (TISE)
- Objective Functions: Minimize TEC, Minimize UTCI, Maximize TISE
- Decision Variables: X1 to X11 (with X8 constrained as an integer)
- Search Space: Dataset bounding box with 20% expansion

Parameters (High Exploration Configuration):
- Generations: 200
- Population Size: 171
- Crossover: Probability = 0.8, Distribution Index = 40.0
- Mutation: Probability = 0.15, Distribution Index = 10.0
- Duplicate Elimination: Enabled (to prevent premature convergence)

Output:
- Pareto Front (Optimal combinations of design variables)

```
START
 Initialize Das-Dennis reference points and a bounded random population.
 FOR generation = 1 TO N_GEN DO:
     Apply Simulated Binary Crossover  to generate offspring.
     Apply Polynomial Mutation to offspring to maintain diversity.
     Enforce variable constraints (e.g., round X8 to nearest integer).
     Evaluate offspring via pre-trained surrogate models:
        Predict TEC and UTCI values.
        Predict TISE value (and negate for minimization framework).
     Merge parent and offspring populations.
     Apply non-dominated sorting to retain 171 individuals.
 END FOR
 Extract non-dominated solutions from the final population as the Pareto Front.
 Export objective values and evolutionary history for visualization.
END
```